

SAKETH PAPAREDDY

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SUMMARY

MS AI graduate (UNT, May 2026) with 3+ years of professional experience building AI products at an IT consulting firm and hands-on expertise in end-to-end LLM application development. Delivered chatbot and NLP solutions across finance and enterprise domains, and independently built a hybrid RAG pipeline (FAISS + BM25 + cross-encoder reranking) with 17× latency improvement, a 6-class query router, and hallucination filtering via n-gram grounding. Focused on applied AI systems — from prototyping and evaluation to reliable deployment at scale.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Pandas, NumPy, Matplotlib, Scikit-learn

ML / Deep Learning: TensorFlow, PyTorch, XGBoost, CNNs, RNNs, LSTMs, ARIMA, Feature Engineering

Generative AI & LLMs: LangChain, FAISS, BM25, Sentence Transformers, Groq API, Llama 3.3 70B, Mistral-7B (4-bit), Prompt Engineering, RAG

Deployment & Tools: FastAPI, Flask, Streamlit, Git, GitHub, Jupyter, Apache Spark, Hadoop

Cloud & Certifications: Microsoft Azure (AZ-900 certified)

WORK EXPERIENCE

Helson Software Solutions Pvt. Ltd.

AI Engineer

Hyderabad, India

Feb 2022 – Jun 2024

- Developed chatbot solutions for 3–4 finance and business clients using OpenAI GPT-3.5/4 API and early LangChain, building prompt chains and conversation memory to handle domain-specific client queries.
- Integrated semantic search using FAISS and sentence-transformers for document retrieval; fine-tuned Hugging Face models for text classification tasks across fintech and enterprise client projects.
- Deployed AI models as REST APIs using FastAPI and Flask, applied prompt engineering to improve output quality, and iterated on model behavior based on client feedback across multiple deliveries.

Python Intern

Apr 2021 – Feb 2022

- Wrote Python scripts for data cleaning, preprocessing, and automation tasks supporting the software development team across multiple client projects.
- Assisted in building and testing REST APIs using Flask, debugged Python codebases, and contributed to internal tooling used across client engagements.

PROJECTS

Generative RAG Pipeline with Hybrid Retrieval & LLM Generation

Python, LangChain, FAISS, BM25, Groq API, Llama 3.3 70B, Mistral-7B, Streamlit | *Live Demo:* saketh-papareddy-rag.streamlit.app

- Built and deployed a production-grade PDF Q&A system on Streamlit Cloud — end-to-end ownership from prototyping through evaluation to live deployment. Uses Llama 3.3 70B (Groq API) with Mistral-7B local fallback, reducing response latency from ~45s to 2.6s (17× faster).
- Implemented hybrid retrieval (FAISS semantic search + BM25 keyword search + cross-encoder reranking) and a 6-class query router built with prompt engineering — 90% classification accuracy, 100% out-of-scope rejection rate.
- Designed and ran structured evaluation on 48 questions across a 176-page technical document; applied n-gram overlap grounding for hallucination filtering, achieving F1=0.51 with zero hallucinated citations on grounded responses.

Retail Store Sales Forecasting Engine

Python, ARIMA, LSTM, TensorFlow, FastAPI, Flask, Pandas

- Built and served two forecasting models — ARIMA (RMSE: 0.0722, directional accuracy: 78.57%) via FastAPI REST API and LSTM (RMSLE: 0.2892, directional accuracy: 67.48%) via Flask — both accessible via web interface without writing code.
- Built an end-to-end feature pipeline ingesting 5 external signals (oil prices, holidays, promotions, store metadata, transactions) with N-step forecasting and CSV export endpoints.

Airbnb Destination Recommendation System

Python, XGBoost, Gradient Boosting, Flask, Pandas, Scikit-learn

- Trained XGBoost on 213K+ user records to rank top-5 destinations (NDCG@5: 0.927), served via Flask web app; ranked top 15% on Kaggle.
- Engineered 500+ features from session logs, demographics, and device signals; trained a second Gradient Boosting model (NDCG@5: 0.824) on booking-only users to compare behavior across segments.

EDUCATION

University of North Texas

Denton, TX

Master of Science in Artificial Intelligence — May 2026

Relevant Coursework: Deep Learning, Machine Learning, Neural Networks, Computer Vision

NBKR Institute of Science & Technology

Vidyanagar, A.P., India

Bachelor of Technology in Information Technology

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900)